

L1.4 The Derivative

Recall: L1.3 Continuity

Name _____

Solo – 9 minutes

1. Is each function continuous at the marked point? If not, name the kind and say **what died** — the limit, the value, or the match. **One of these three is continuous.**

(a) $\frac{x^2 - 4}{x - 2}$ at $x = 2$

(b) $\frac{1}{(x - 1)^2}$ at $x = 1$

(c) $|x - 3|$ at $x = 3$

2. $g(x) = \frac{x + 3}{x^2 - x - 6}$. Where is g continuous?

Then say how you knew — **without taking a single limit.**

3. For $f(x) = x^3$, find the slope of the line through $(1, f(1))$ and $(1 + h, f(1 + h))$.

h	$1 + h$	$f(1 + h)$	slope
1	_____	_____	_____
0.1	_____	_____	_____

What number do you think these slopes are heading for as h gets smaller?